



Wireless body area network (WBAN): A review of schemes and protocols

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ARTICLE INFO

Article history:

Received 31 October 2020

Accepted 26 May 2021

Available online 23 June 2021

Keywords:

WBAN

WSN

Energy efficiency

Fault tolerance

Routing and medium access

ABSTRACT

To provide practical solutions to the difficulties faced by current available medicinal service based frameworks, Vital information of patient ought to be conveyed to the concerned destination continuously in real time by means of the web which gives rise to need to a system which allows any patient to get monitored over internet with the help of a network called as Wireless Body Area Network (WBAN) comprises of several sensor nodes that consume low power as compared to other sensing devices.. The utilization of these networks includes the assistance of human safety in various situations. Though WBAN is extremely useful in today scenario, several challenges need to be tackled before its deployment. This article provides a detailed study of various schemes and protocols used to deal with energy efficiency, fault tolerance, security and privacy. The study in this article also identifies challenges that need to be considered before the design and development of any protocol for WBAN.

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Selection and peer-review under responsibility of the scientific committee of the National Conference on Functional Materials: Emerging Technologies and Applications in Materials Science.

1. Introduction

Wireless broadcasting has been playing a very crucial role in our daily life from the past few decades. It contributes in several ways since it is an essential part of many practical applications like tracking the devices, tracking appliances, monitoring, and Automation. The world is running into a state of rapidly growing population due to which the average life expectancy has also increased, especially in developed countries like Japan, Spain, Germany, and Italy, etc.

As there is a rise in the population, average number of people ageing 60 and above is also increasing, who need more attention towards their physical health, thus incurring more expenditure on medical treatments. Unfortunately, fatal and chronic diseases like cardiovascular, cancer, and asthma, etc. diseases are often diagnosed very late, due to which it increases the average number of death rates of the individuals. If we can detect these diseases at an early stage, we may be able to mitigate their effect and increase the sufferer's life expectancy [1].

Smart healthcare has gained a lot of attention owing to its merits like mobility, flexibility, and ease of monitoring of the patient. It is popular since it makes use of a variety of latest technologies like cloud computing, IoT (internet of things), and big data processing, which focus on building an advanced disease care platform. It practically involves sensing technologies and automated devices to

perceive and identify the physical properties of the patient and then perform several calculations, process those calculations, and mines the knowledge through those calculations, which helps us to cure the patient more efficiently than the traditional monitoring systems [1].

2. Wireless body area network (WBAN)

WBAN is a kind of Wireless Sensor Networks which consists of small bio-medical devices abbreviated as nodes, committed to ensuring continuous monitoring of patient via some vital parameters. WBAN comprises of low-power components that act inside or across a human body to support numerous applications, such as in medical applications. While WBANs and WSNs share several challenges, including miniaturization, reliability, security, etc. The primary research issues remain in the framework for tracking healthcare and bio-medical devices. Undoubtedly, the advancement of WBANs ought to follow the expanding improvement in the medical field. The fundamental goal is to ensure that patients are checked at home so that to avoid late diagnosis [2].

The use of a WBAN gives the power to monitor continuously one's vital parameters; therefore, it offers greater flexibility and mobility to the patients. Importantly, Doctors will be having a broad and clear view of the patient's current reports. As the name suggests, WBAN's are the network associated with the human body

where several of its parts are on, above, or inside the human body. These parts are generally abbreviated, as shown in Table 1.

The architecture of WBAN is depicted in Fig. 1. The WBAN can be a particular form of a network built and established for the physical body to track, control, and communicate various vital physical body signs such as temperature, vital signs, ECG, etc. Such vital signs are also monitored using different sensors that are mounted on the body or on clothes that may be under the human skin. Many types of medications are also delivered with actuators to monitor the vital signs of human beings. It is the central unit is responsible for deciding wireless contact between actuators, sensors, and CCU. The fundamental unit of the person is used to receive all information from a human body to the outside world (emergency, doctor). WBAN is another form of personal wireless network and around the physical body where each node is assigned a logical address. IEEE standard 802.15.6 was assigned to build and upgrade energy-efficient devices for WBAN applications. Table 2 represents the specifications [3,4]:

2.1. Type of devices used in WBAN [4]

Wireless sensor node: It is responsible for collecting data on a physical stimulant and, if necessary, processing the data and reporting it through wireless transmission media. It includes multiple components such as sensor hardware, processor, memory, power unit, and transmitter/transceiver.

Wireless actuator node: The actuator nodes responds to the data received by a sensor. The actuator components are identical to sensor components: actuator hardware (e.g., medicine control hardware, including a medicine bank), power generator, processor, memory, transceiver, or receiver.

Wireless personal device (PD): An external portal, monitor, actuator, or display/LEDs support device on the screen holds the complete information gathered by the actuators, sensors and transmits the same information to the user. A large processing unit, memory unit, power unit, and transceiver are the main components.

2.2. Devices used in and their data rates

WBAN consists of several heterogeneous devices, so their data requirement and bandwidth varies according to their specifications. Some of them are listed in Table 3 and Table 4.

Table 4 represents the accuracy rate, data rate, and bandwidth of various applications used in WBAN.

2.3. Standards and techniques used in WBAN [6]

IEEE standard 802.15.6: The IEEE working group has several success stories in the realization of the WBAN universal standard of communication. IEEE 802.15.6 Working Group shall in April 2010 create its first draft communication protocol for WBAN. It has been designed for different medical and non-medical applications for in body/on-body nodes. In February 2012, the accepted version for the same was amended. The goal is to establish and function on, in or around the human body (but not limited to humans) a communication protocol for low power devices to serve a number of

applications including medical, consumer electronics, personal entertainment, and others.

Requirements of WBAN following IEEE 802.15.6 are as follows:

- (i) Bit rates of between 10kbps and 10mbps should be allowed.
- (ii) Packet Error Rate (PER) for a 256-octet payload should be <10 per cent.
- (iii) Nodes in the network should be able to get added and removed to the network in only 3 s or less.
- (iv) WBAN should be able to support up to 256 nodes.
- (v) Nodes should be able to communicate reliably even if the body is moving.
- (vi) Latency should be <125 ns in medical and 250 ns in non-medical applications.
- (vii) There should be a well-known power saving mechanism.
- (viii) It should have incorporated the QoS management system.

IEEE standard 802.15.4 (ZigBee): IEEE 802.15.4 is one of the protocols which consumes minimum power and used by wireless sensor network applications with limited data rates. It consists of many frequency bands: 868 MHz, 915 MHz, and 2.4 GHz. A total of 27 sub-channels are assigned in IEEE 802.15.4, i.e., 16 sub-channels in band 2.4 GHz, 10 sub-channels in band 915 MHz and one post-channel in band 868 MHz respectively. MAC Layer operates on several layers, such as Physical, Data Link Layer, Network Layer, and Transport IEEE 802.15.4, which is part of the data link layer

The frame structure is bounded by various channels, as shown in Fig. 2.

IEEE 802.15.4 Standard generally works on security module in WBAN which covers following security requirements of WBAN:

Access Control: Information of every user in the WBAN network is vulnerable, so there should be proper access control mechanisms to secure the information available on the server.

Confidentiality: According to this parameter, there should be a limited number of authorities, or we can say there must be some legitimate users who should have a view on the information available on the server.

Frame Integrity: Data should be lossless, i.e., whatsoever we are transferring with the help of WBAN should reach the destination server accurate and complete.

Sequential Freshness: Since all wireless sensor networks have some form of time-varying measurements, it is imperative to ensure that analysis is recent and updated.

Short Range Radio Frequency (R.F.): Frequencies to be used by WBAN are maintained by the communication authorities, which may differ in various regions and countries [7]. Some of the short-range frequency bands used in WBAN are as follows:

Medical Implant Communication System (MICS) Bands: MICS is a low-power short-range communication network with a high data rate. Frequency Range is 402–405 MHz for the same. Data transmission to facilitate diagnostic and therapeutic functions related to medical implant devices has been globally recognised.

MED Radio Frequency Band: The medical device radio-communication service is in the range of 401–406, 413–419, 426–432, 432–438, and 451–457 MHz. These kinds of Spectrums are used to diagnose implanted as well as a sworn body device.

Wireless Medical Telemetry Service (WMTS) Bands: The wireless medical telemetry service is in the range of 608–614, 1395–1400, and 1427–1432 MHz. These kinds of spectrums are used to monitor the patient's health remotely. With the help of these bands, we can transfer data (Vital information of the patient) to a remote location, Such as a nurse's station, Equipped with a specialized radio receiver [8].

Industrial Scientific and Medical (ISM) Bands: These bands generally uses 2.4 GHz spectrum for wireless communication. It includes

Table 1
Sensors and its functionalities [2]

Parts	Functionalities
Implant node	They are positioned below a sink node in the human body.
Body surface node	It is located on the surface of human skin.
External node	No interaction takes place with the skin.

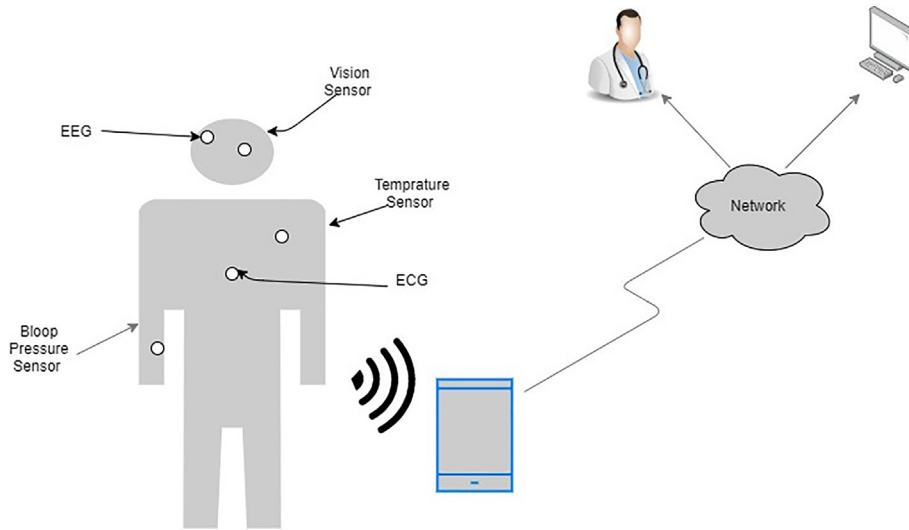


Fig. 1. General view of WBAN [3]

Table 2
WBAN specifications.

Attributes	Values
Start-Up Time	<100 ns
Network set up a time	<1 sec per device
Distance	In standard case 2 m and in special case 5 m
Network Density	2–4 nets/m ²
Power Consumption	<1mW/Mbps
Network Size	10 ns
Latency	Max 100

Table 3
Functionalities of WBAN [5]

Devices Used	Functionalities	Diagnosis
Accelerometer	To monitor the speed of different nodes in WBAN	Supplementary data
ECG	To track heart activities	Cardiac attack
EMG	To detect muscle function activities	Disorder of muscles and nerves
EEG	Monitors brain-related activities	Epilepsy
Blood Pressure Sensor	To measure the force exerted by circulating the blood on the wall of blood vessels	Hyper-tension
Gyroscope	Monitors orientation and angular velocity of nodes in WBAN	Supplementary data
Pulse Oximetry Sensor	Tracks saturation level (SpO ₂) in the patient's blood	Lung Cancer, Asthma
Respiration Sensor	Monitors breathing-related activities of human beings	Loss of consciousness

Table 4
Data rates for different medical applications [5]

Sensors Used	Data Rate	Bandwidth
ECG	71kbps(6 leads)	100–500
	288 kbps(12 leads)	100–1000
EMG	320 kbps	0–10000
EEG	43.2 kbps	0–150
Glucose Monitoring	1600 bps	0–50
Blood Saturation	16 bps	0–1
Motion Sensor	35 kbps	0–1
Temperature	120 bps	0–500

Qualified appliances designated to generate and handle local radiofrequency for Industrial, Scientific, Medical, and domestic purposes in the field of telecommunications.

Bluetooth Low Energy (BLE) and Bluetooth: Bluetooth is another wireless communication standard operating within the ISM 2.4 GHz frequency band in the range 2400–2483.5 MHz. Data is split in the form of packets and exchanged via Bluetooth channels (having bandwidth equal to 1 MHz). The difference between low-energy Bluetooth and Bluetooth is just in energy consumption. In 2011, BLE was entered in the market as Bluetooth 4.0. Like Bluetooth, BLE also works on an ISM band of 2.4 GHz. Unlike the Bluetooth standard, BLE remains in sleep mode, except when contact is formed. Fig. 3 shows the Different band frequencies we can use in WBAN.

2.4. Applications of WBAN

WBAN applications are spread into a variety of domains such as Entertainment, Military, Ubiquitous health care, Sports, and many others. The main area for the same is how they put an impact on improving the quality of life. Technical requirements are generally application-specific in WBAN. Fig. 4 shows the different categories of WBAN applications:

2.4.1. Medical applications

Medical applications allow us to monitor someone's physiological attributes as Heartbeat, blood pressure, body temperature. In case of any unusual data detection, it should be collected by sensors. It will be transferred to a gateway, which will further forward that data to a remote location (Doctor's room) via the internet so that necessary action can be taken. These applications are further categorized into three sub-categories as follows:

■ Wearable WBAN [10]

- (i) **Soldier Fatigue and Battel Readiness:** Some applications are used to monitor soldier's activity in battel field more closely. Majorly these WBAN employs GPS (Global Positioning System), biometric sensors, cameras along with an integration unit, and a centralized control network for contact with other soldiers.
- (ii) **Professional Sports Training:** Training modules for the athletes can be quickly scheduled and planned as WBAN applications can help us to monitor their motion capture and rehabilitation.

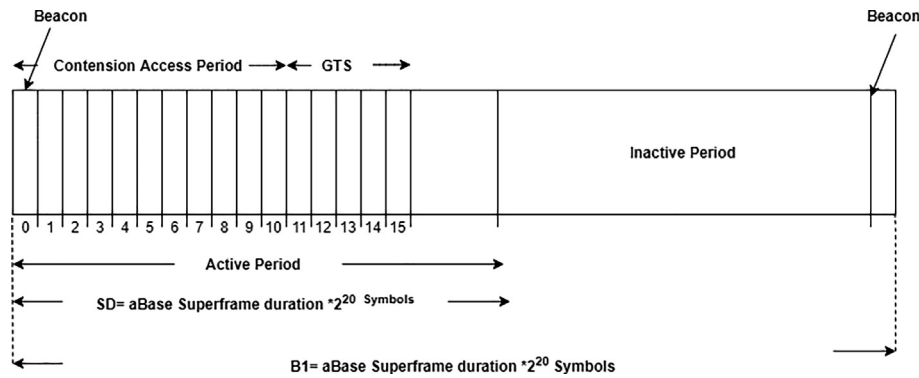


Fig. 2. Superframe Structure [6]

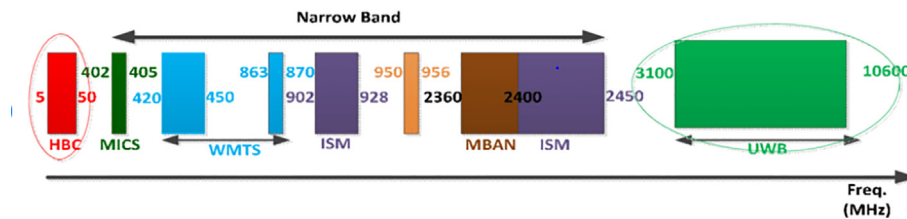


Fig. 3. Radio Frequency Spectrum for WBAN.

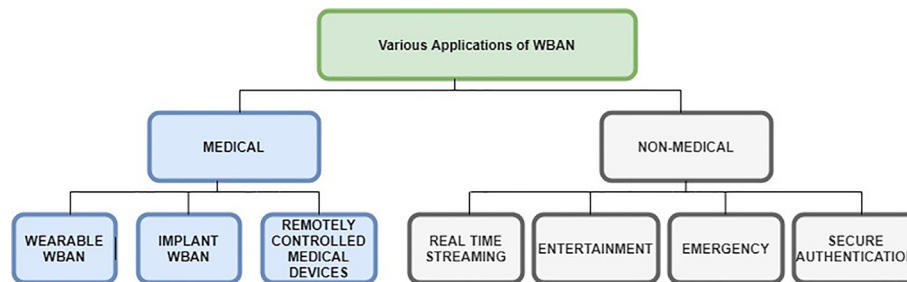


Fig. 4. Various Applications of WBAN [9]

- (iii) **Sleep Staging:** Sleep disruption can be conducted via a polysomnography test that involves the examination of various bio-potentials collected in a sleep laboratory overnight. The study includes numerous cables running from the centre to a box attached to the patient device, which can distract from sleep disruption.
- (iv) **Asthma:** A WBAN, together with various sensors, can also be used to track allergic diseases in the air and provide a physician with real-time feedback, which can benefit millions of asthma-patients.
- **Implant WBAN:** Implant WBAN is comparable to nodes that are located either under the skin or in the bloodstream in the human body. Many of the leading implementations are as follows[11]:
 - (i) **Diabetes:** According to medical science, diabetes is a kind of disease that never happens in a short time duration; it always is a result of long term medical history and ignorance of the patient. The regular monitoring supported by WBANs decreases the risk of fainting, ensures proper dosing, and reduces the risk of circulation loss,
 - (ii) **Cardiovascular Diseases:** The monitoring of episodic occurrences and other irregular conditions through WBAN technology will significantly reduce myocardial infarction (MI).
 - (iii) **Cancer Detection:** WBAN-based sensors are competent for controlling cancer symptoms in the human body would enable the physician to continuously diagnose tumours without any advanced treatment providing more timely diagnosis and care.
- **Remotely Controlled Medical Devices:** The continuous monitoring of patients from a remote location and providing various facilities is abbreviated as Ambient Assisted Living so that every WBAN connects via a medical back-end network. AAL seeks to improve the self-care of patients. The same is true of any of the following[12]:
 - (i) **Patient Monitoring:** Some of the main applications of WBANs needs monitoring vital signals from the human body, along with offering feedback and valuable data about the rehabilitation process in applications that controls health. In particular, it detects vital signal indicators, for example, body temperature, blood pressure, heart rate, breathing rate, parameters of body implants, and wireless chest sounds.
 - (ii) **Tele-medicine System:** Tele-medicine systems that use a power-intensive protocol such as Bluetooth, would be vulnerable to hindrance from other devices functioning at a specific range, or wireless bands allocated for conveying information to faraway stations. And they are limiting con-

tinuous surveillance. Whereas incorporating WBANs into a telemedicine program enables unobtrusive outpatient health monitoring for long periods.

2.4.2. Non-medical applications

- (i) **Real-Time Streaming:** The Streaming involves buffering of images, such as recording a video clip in a mobile phone, and even 3D images. Audio streaming is also one of these applications' functions. These applications can also be used to recognize body movement, body-based entertainment, and vital sign, recognize, detect emotions, and the track was forgotten items by sending a warning to the person.
- (ii) **Entertainment:** Entertainment consists of social networking applications and games. We may also include various devices such as MP3 players, microphones, WBAN head-mounted displays. Virtual reality is also one of these applications' popular sector.
- (iii) **Emergency:** Any of the implementations may also be used to detail emergencies, such as home fire or household flammable/poisonous gas, and to transmit this data to the person with body-worn devices to avoid unnecessary damage.
- (iv) **Secure Authentication:** In such applications, we examine biometrics in both biology and behaviour, for instance, an iris recognition, facial patterns, and fingerprints. It is one of the WBAN's critical applications for forgery and duplicability.

2.5. Faults and failure in WBAN [13]

To understand the mechanisms of fault tolerance, a person needs to keep in mind the difference between error, mistakes, and failures. Specific meanings of the terms were used.

- The fault is some form of defect that results in an error.
- An error relates to the device's incorrect (undetermined) condition. That kind of state can lead to failure.
- Failure is evidence of a manifest error, which arises due to deviated output by a node from its specific value.

2.6. Sources of faults and failure in real WSN [14]

WSN are extensively deployed under extreme conditions and is vulnerable to faults in several layers of the network. Fig. 5 can be viewed as a layer-wise architecture of a wireless network of sensors to determine the faults that arise in real usage.

- **Node Failure:** Nodes have many software and hardware components which could cause system failures. Node failure can occur due to different causes, such as:
 - (i) The nodes of the sensor are in direct contact with water because of which short circuit can occur.
 - (ii) Once the node's battery hits a certain amount of sensor reading, it can be bogus.
 - (iii) Some hardware failure can also cause node failure.
- **Sink Failure:** The sink device collects and circulates the data obtained on the network and subjected to component flaws in back-end system at a higher network level. If the device fails, a significant network failure occurs unless fault-tolerant measurements are available, because the data on the sensor node can not be processed.

These devices can be installed in places where there is no power source available; usually, batteries are added along with solar cells to provide the necessary amount of energy in these applications. In

the glacial expedition, this modern approach proved to be ineffective.

- **Network Failure:** Routing is one of the foundational blocks within a WSN. It is necessary to collect sensor data, distribute updates to the software, and configuration and coordinate between nodes. Additionally, application-specific routing protocols can be required to track moving objects and "follow them." Faults on the network layer can lead to messages missed or lost, and unexpected delays.

3.

Literature Review: Over the years, many of the researchers have worked upon various challenges in WBAN some of the major challenges are as follows:

2.7. Energy efficiency in WBAN

Zahid Ahmed et al. [15] suggested an energy-efficient clustering and cooperative routing protocol for WBAN. E-HARP typically uses two mechanisms to enhance network efficiency, i.e.,

- (i) Selection of dynamic cluster heads for cluster members,
- (ii) The mechanism to lower down the transmission of redundant data sensed during transmission by sensor nodes.

In the protocol, energy recycling is used will supply the network's electricity countlessly by using resources across the human body such as solar energy along with dual sinks to support the network's path-loss problems. The scheme follows the process to pick a cost of cluster head that includes multiple factors such as transmission capacity, residual energy, etc. The author has found that the proposed method outperforms better than other techniques.

The work in R. Priyanga et al. [16] proposed an M-PSK based Energy-efficient WBAN. The author used two schemes in work named as Rate optimized and Optimum Energy Scheme with received Diversity. The author has also used the EEWBANRD algorithm to improvise energy Efficiency in WBAN. The author has also proposed that along with energy efficiency; this method can also work well to improve different factors like overall delay, Throughput, and network lifetime in WBAN.

The work in Ameneh Rostampour et al. [17] proposed a novel topology design that reduce the energy consumption and overall cost of WBAN. The proposed method considers the relay nodes and sensing nodes along with their position and number in WBAN. An algebraic topology model called the energy-conscious WBAN design model (EAWD) is used to enhance the number of relay nodes and their location. The proposed topology design specifies the optimum replacement of relay nodes and in addition to the routing of traffic and also how traffic can be divided into multi-paths to reach the sink node.

Khanna et al. [18], proposed a routing protocol named as an energy-efficient protocol in WBAN. The proposed method has used the group of single and multi-hop transmission and has compared the same with the only multi-hop transmission. In every round of data transmission, the node was decided based on the cost function, and a proposed approach has used a new forwarder or a parent node. Results of the experiments revealed that the protocol has improved packet transmission to minimize overall efficiency and system reliability time by 70per cent and 35 per cent, respectively.

In Guangdong Yang et al.[19] an energy efficient protocol for WBAN is developed. An analysis and simulation findings show that the protocol will take the difficulty of replacing the sensor node in human bodies into account; thus, our protocol greatly increases the lifespan of the network and fulfils the sensing data hardware requirements in the case of an emergency.

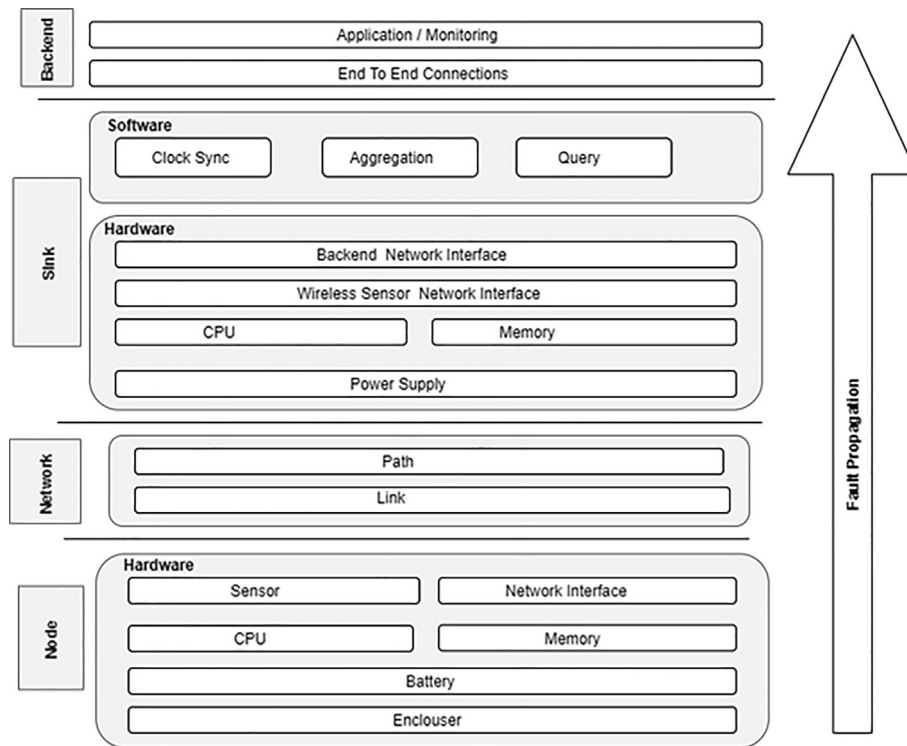


Fig. 5. Fault Classification and Propagation [14]

2.8. Fault tolerance in WBAN

Yuhuai Peng, Xiaojie Wang et al. [20], proposed a mechanism for smart Healthcare Systems based on an effective network coding fault in WBAN has been proposed. The author has tried to resolve the limitation of previously developed remote healthcare systems by incorporating the Internet of Things definition within it. Cloud computing and big data were used to prevent the failure and delays of unnecessary data.

The objective of A. S Albahri et al. [21] is to enhance the system for continuous healthcare of patients with coronary heart disease. The author has adopted a 3-tier architecture in which tier-1 involves the introduction of Healthcare Continuous Integrated Congestion Avoidance and Control (HOCA) Protocol. It deals with end-to-end delays and unnecessary congestion energy consumption, and tier 2 has been generally divided into three major sections where first deals with treatment support and second-part disease tracking deals with Triage over mhealth and the third part deal with triage-based disaster accidents and casualties. At the end, tier 3, protection and privacy are dealt through the implementation of low cost and stable (LCS) communication framework. The author has accomplished the change in the Internet of Things by implementing a 3-tier adaptive system known as FTF- mhealth-IoT by using all three-tier rates.

2.9. Routing protocols in WBAN

(i) posture based routing:

Karmakar et al. [22], proposed an enhanced a routing protocol named MHRP. This technique is specifically designed for the acquisition of heart rate where seven nodes are situated on the human body. The author has suggested a new form of fault-tolerant framework that would consist of two similar and separate node sets.

Samanta et al. [23], The Author has presented an efficient and distributed opportunity routing NCMD which produces the mini-

mum network management costs. The author has done certain experimental work indicating that WBAN is dynamic in nature, the configuration of the network is both in an irregular and continuous state, which can cause the great transmission latency and increased costs of network maintenance. In fact, with the rise in mobility level, different service quality (QoS) indicators in the network are getting worse each day. An opportunistic transmission connection establishment algorithm was used to solve the mentioned problems, to reduce data management costs and maximize the Qos.

(ii) temperature based routing:

Tang et al. [24], has introduced a thermal-conscious routing protocol that takes temperature as a metric for data transmission route selection and selects the neighbour node with the optimum temperature from other nodes in the network as its next step. The retreat strategy is followed when data experiences a hotspot in data transfer to restore the original data route to find the right route. The back-off strategy protocol is used to acquire good results, and also gave suggestions for future study.

Bhangwar et al. [25], proposed thermal aware and trust-based routing protocol (TTRP). Through the work, the author has presumed two confidence and temperature variables for the study of the routing protocol. As compared to other routing techniques, this routing approach provides an additional relay node to obtain and transmit data that it does not engage in the collection of information and is designed with maximum energy to accomplish the function of conveying information from several nodes.

(iii) Cluster-based routing:

Ullah et al. [26], suggested a dual sink node routing protocol DSCB based on cluster and uses dual sink nodes. Before the above approach, authors have noticed several shortcomings in WBAN routing protocols viz the high data transmission

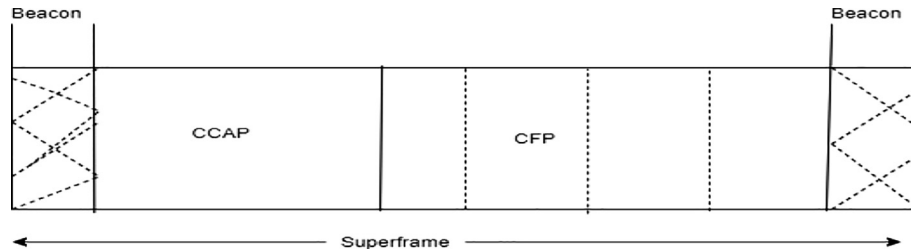


Fig. 7. Superframe structure.

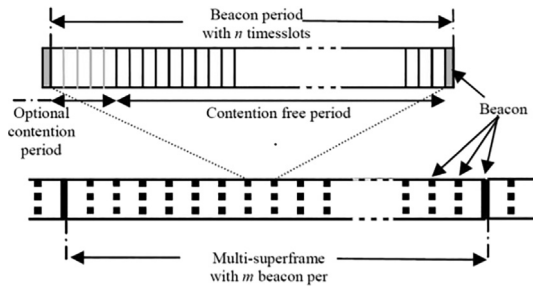


Fig. 8. Superframe structure for MedMAC Protocol.

Table 5
Summary of proposed work.

Reference	Issue	Limitations
[15]	Energy Consumption	The author have not considered Overall delay, the accuracy of transmitted data over the network
[16]		Overall delay increases along with no effect on network life.
[17]		The user has done no work on security and authenticity. Data loss is that significant drawback
[18,19]		Many major factors like overall delay, security and data loss is yet to be considered here.
[20]	Fault Tolerance	Overall energy consumption still remains an issue here, and there is no mechanism which ensures the network lifetime and authenticity of data flowing in the network.
[21]		Overall security and data loss remain a big issue in this work.
[22,23]	Routing Protocol in WBAN	The authors worked upon improving the routing mechanism in a small amount of data, whereas a big amount of data is still a concern for both to take care.
[24,25]		In TARA sudden temperature rise is one of the major problems whereas in TTRP discarding mechanism has been a concerning issue.
[26,27]		Both proposed methods have worked on sink nodes in the network, but packet loss is still an issue which needs to be taken care.
[28]		The major concern with respect to this protocol is energy consumption and packet formation in the network layer.
[29–32]	MAC Protocols in WBAN	All listed protocols have worked upon the network traffic and network lifetime. Still, there are many other challenges which can improve like, reliability of packet during transmission, improving average delay factor and energy efficiency.

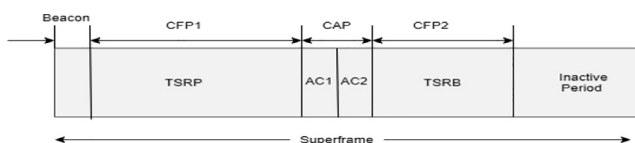


Fig. 6. Superframe Structure for Priority Guarantee MAC [33]

loss, congestion and restricted coverage. The proposed technique helps in improving the performance of the WBAN routing protocol about transmission delay and network lifetime via dual sink node.

(iv) quality of service based routing:

Vitale et al. [27], proposed a data-aware HDPR hybrid routing protocol. This method has followed a modular approach. The major focus of the author here is on the data travelling on medium access control (MAC) and network layer. The recipient incorporates the data into the data-link layer and then delivers it to the Data Classification module, which partitions the data into Delay-Sensitive Data (DSD), Regular Data (RD), Essential Data (ED), and RSD. Data preference is high to low ED, DSD, RSD, and RD Via their respective data modules; data is being grouped into the QoS perception module.

2.10. MAC protocols in WBAN

Su.H et al. [28], presented a battery aware TDMA based MAC protocol. The author has used a cross-layer design for the same to enhance the network life in this work. According to the author, the following parameters are to be considered for the better performance of the network, time-varying fading capabilities of the wireless channel, and packet queuing characteristics.

It operates in a similar mode as beacon-enabled IEEE 802.15.4 Mac protocol which works on periodic updates where time parameter will be split into three major parts beacon slot, active and inactive time slots. The author has considered the frame length here as an adaptive parameter that can be changed according to the application requirements. Generally, nodes stay in sleep mode during the inactive time to prevent extra energy consumption. This protocol extends the lifetime of nodes with wireless sensors. This protocol ensures efficient and timely delivery of the packets.

Zhang y et al. [29], proposed a priority guarantee MAC protocol. According to this work, this protocol follows a new superframe structure, as shown below in Fig. 6. The active period in this protocol is partitioned into five major parts a beacon, Control channel 1 (AC1), Control channel 2(AC2), period reserve for periodic traffic (TSRP) and time reserve for bursty traffic (TSRB). AC1 is often used to uplink life-critical medical applications control and AC2 can be used to bind consumer electronic appliances. It randomizes Aloha protocol which allocates guaranteed time slots (GTS) to TSRP and TSRB. The author has also claimed that this protocol outperforms the efficiency of IEEE 802.5.4 and simple TDMA based Mac protocol. However, emergency data traffic and complex superframe structure are still major issues to deal with.

Kwak et al. [30], proposed a new mechanism at a medium access control layer to ensure reliable transmission. The author has also implemented two separate systems called the traffic-based waking-up mechanism for regular traffic, and another is the emergency or on-demand radio wake-up mechanism. Sensor nodes produce regular traffic to recognize physiological signs of

the body wherein emergencies body sensors provide warning signals. In [10], the author has proposed a new superframe structure. The suggested approach favoured slotted aloha over the random aloha technique. Monte Carlo technique has been used in the proposed system for poison simulation and deterministic flow. In terms of energy consumption and network lifetime, the proposed approach performs better than WiseMAC. Fig. 7 depicts the use of CCAP and CFP in MAC protocols.

N.F. Timmons et al. [31], proposed a medical MAC protocol (MedMAC). The proposed method tried to improve the medium access mechanism, along with the conservation of residual energy. For time slots distribution, MedMAC has used TDMA approach for transmission of data and communication to end nodes. Where these time slots are of varies in length and behave differently according to sensor node needs. The proposed approach uses structure with multi-super frames, where beacons are used for synchronization. However, it utilizes an optimal contention cycle for the initialization of network, emergency traffic, and low-data communication. The proposed approach uses the time-stamp scavenging with Adaptive Guard Band Algorithm (AGBA) to maintain node and coordinator synchronization clockwise. Data packet collision is prevented with the use of separate GTS for each sensor node in the network. Super frame structure for the same use dynamic slots as shown in Fig. 8.

Based upon the above survey, we can say that irrespective of the popularity that WBAN has gained in the past few years. There are several changes we need to cope up with. Table 5 shows a brief analysis of the previous work done on different challenges and their proposed solutions.

3. Conclusion

WBAN has a wide scope in Research & Development in the healthcare sector. With the advancement in wireless sensor networks has a degree of improvement in the healthcare domain has increased. WBAN systems can help in providing a better communication system with infected human beings to acquire their essential information and provide them with the services in less time before the severe danger of diseases threatens their lives. Even though WBAN has the potential to offer a wide range of benefits to patients, in terms of healthcare and telemedicine field, there are many serious issues related to them as well. This paper gives a broad view of some of the critical issues and limitations of Wireless Body Area Network. This paper serves as a call to action for better design in terms of security, privacy, data authenticity, Fault tolerance, routing, etc. Many researchers have done a lot of work in the direction of improving the efficiency and benefits of WBAN, but still, we have a lot to improve.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Further Reading

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